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Instagram Smart Assistant (ISA)

(Project Report)

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Abstraction

This report provides a comprehensive analysis of the Instagram Assistant project, detailing its objectives, features, and related design aspects.

The report begins with an overview of the current system for the Instagram application, then identifies the goals of this project and reviews the system requirements, including both functional and non-functional requirements.

It also clarifies the target audience and use cases through user stories and User journey map.

The design section addresses architectural design and documentation plans, including user guides and troubleshooting solutions.

Finally, the report concludes with recommendations for future work and related references, providing a structured approach to the project's development and implementation.



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Chapter 1-Introduction

1.1 Purpose of the Project

The Instagram Smart Assistant service seeks to improve the user experience and help them achieve the desired results. The Smart Assistant also facilitates interaction with advanced features, which contributes to achieving user satisfaction and improving their experience on the Instagram platform.

1.2 Overview of the Project

This system is designed to help users conduct conversations, respond to messages, and provide suggestions based on pattern analysis.

Key Features:

The assistant includes features such as suggestions based on user interactions, and data analysis to provide advice on better audience engagement and controlling content display algorithms.

System Architecture:

It is based on a web architecture that operates on external servers supported by artificial intelligence technologies and natural language processing (NLP).

User Access Mode:

The smart assistant is accessed through a graphical user interface integrated into the Instagram application.

System Environment:

The assistant operates within the Instagram environment and can interact with all chat and communication features of the application, without requiring any special conditions or additional requirements.

1.3 Related work (literature review)

- 2 In this chapter, the technologies and applications similar to the Instagram smart assistant will be discussed, along with a comparison and clarification of the differences between it and those technologies.

1. Google Assistant:

Google Assistant is one of the most widely used smart assistants, offering features such as intelligent search, voice control, and integration with smart home devices.

It shares with the Instagram smart assistant the ability to respond to voice commands, accelerate access to content, and provide assistance to users based on context or problems.

The difference between it and the Instagram smart assistant is that the smart assistant technology is more focused on enhancing the user experience specifically on social media platforms, particularly Instagram, unlike Google Assistant. The Instagram assistant can provide more precise tips or suggestions related to social interaction and communication, and it utilizes content filtering and analysis algorithms.

2. Canva Assistant:

Canva Assistant provides intelligent assistance in designing visual content, such as format suggestions, automatic adjustments, as well as answering user inquiries and solving problems they encounter.

The similarity between them lies in providing suggestions to enhance the user experience; however, Canva Assistant focuses only on designs and images, and the data it handles is limited. In contrast, the Instagram smart assistant can analyze and aggregate a broader range of content, extending beyond mere

post design to include suggestions for interacting with and sharing content. Additionally, it processes a significantly larger amount of data, making it more comprehensive and complex.

3. ChatGPT:

ChatGPT is an intelligent conversational model that can answer various questions, engage in dialogue, and provide diverse assistance.

It shares with the Instagram smart assistant the ability to process text and respond to user inquiries.

While ChatGPT is general and offers assistance on a wide range of topics, the Instagram assistant is more specifically designed to meet the needs of Instagram platform users.

1.4 Purpose of this Document

The purpose of this document is to provide a technical and detailed description of the features and design of the “Smart Assistant” project in Instagram. The document includes clarification of the project’s requirements, along with a description of the overall program structure and the technical design adopted by the assistant. It also highlights the artificial intelligence and machine learning technologies used in data analysis and recommendation generation.

1.5 Overview of this Document

This document includes several main sections covering the project goals, system analysis, architectural design, and implementation strategies. It also provides guidelines on how to use the smart assistant effectively, as well as useful resources for both users and developers. By organizing the content logically, the readers can understand the core components of the project and how they interact with each other.

1.6 Existing System

1.6.1 Existing system description

Currently, the Instagram app relies on a set of basic features that allow users to interact with content, but the experience remains insufficiently personalized, making it difficult for users to discover new content that matches their interests. The current interface lacks intelligent suggestions that could enhance the users’ experience and boost their interaction with the app.

1.6.2 Problems in the existing system

Users face difficulties in finding new content that aligns with their interests, in addition to their limited interaction with the available features. The app’s interface suffers from a lack of customization, necessitating comprehensive improvements in how users interact with the app. This is where the need for the “Smart Assistant” arises to meet these needs and enhance the user experience.

1.7 Summary

The Instagram Smart Assistant project aims to improve user experience and enhance engagement using advanced, AI-powered intelligent features, which increase user satisfaction and interaction.

The features include automatic suggestions based on user interactions, automated responses to messages, and data analysis to recommend content that best matches user interests.

The target audience includes all Instagram users without discrimination, aiming to provide a neutral and personalized experience that suits the diverse needs of users.

The current Instagram system’s drawbacks include a lack of personalization, which makes it difficult for users to discover content aligned with their interests. The project seeks to address this gap through the Smart Assistant feature.

This report includes the project objectives, requirements, architectural design, as well as user guide instructions and assistance tools to ensure the ease of use of the Smart Assistant.

Chapter 2-System Analysis

2.1 Software Features

The ISA includes features such as automatic responses to direct messages, suggestions based on user interactions, and data analysis to provide advice on better audience engagement and controlling content display algorithms.

2.2 Intended Audience

The ISA serves all Instagram users, regardless of their ethnic, gender, or religious differences, and aims to provide a fully personalized and neutral experience, making it a suitable tool for all users worldwide.

2.3 Documentation Assumptions

The smart assistant on Instagram will enhance the user experience by providing interactive and personalized support, such as suggesting post ideas, filtering content appropriately for children, offering content suggestions that match the user's interests, and supporting text and voice.

2.4 Validate the Assumptions

The initial results of implementing the smart assistant technology are promising. User feedback on the program has shown an improvement in the alignment of content suggestions with users' interests, thanks to the AI algorithms that analyze user activity and provide appropriate content based on that. The content filtering feature for children utilized previous content analysis algorithms to identify content containing violent or inappropriate scenes to block within the app, in addition to suggesting content that matches children's interests.

2.5 Software Requirements

2.5.1 Major System Requirements

- System must be available on the Internet
- System must be available 24 hours per day
- System must be accessible by mobile (iOS - Android) and desktop devices.

2.5.2 External interface requirements

1- User interface requirements

- Access Button: Provide a dedicated button for accessing The Smart Assistant in a visible location within the chat area of the app.
- Text Interaction: Users can type and interact with the assistant by entering commands as text. Responses will appear in a chat window.
- Voice Interaction: In the chat area, voice recognition technology is available to understand user commands.
- Image Recognition: Attachment icon for sending images or files.
- Notifications and Tips: Users can enable the feature to receive visual notifications or tips when needed.
- Interactive Guidance: When a new feature or setting is activated, the assistant provides instructions to explain how to use these features.

2- Hardware interface requirements	Supported Devices: - Smart phones, tablets, and desktops. - Supported Operating Systems: All major operating systems, such as iOS and Android. - Compatibility with Wearable Devices: Supports wearable devices like smart watches. Network Requirements: - Internet Speed: The smart assistant works on minimum internet speeds (3G and above), but for a smoother experience, 4G or Wi-Fi connections are recommended. - Multiple Networks: The smart assistant can adapt to a variety of networks to avoid service interruptions or delays.
3- Software interface requirements	Front-end: - Framework: Utilize a framework to develop an interactive and user-friendly interface. - Interactive Libraries: Libraries for efficiently fetching data from back-end servers. Back-end: - Framework: Develop the back-end servers that handle user requests and manage Instagram data. - Databases: Utilize Instagram databases to store data related to user preferences and analytics. - Communication with Instagram: Use the Instagram API to fetch user-related data such as posts, comments, and likes. AI Libraries and Technologies: - Natural Language Processing (NLP): Use specialized libraries for processing voice commands and text analysis. - Machine Learning: Utilize services to train models that predict content preferences and provide recommendations to users. - Recommendation Algorithms: Develop algorithms based on analysing users' previous activities to suggest the most relevant content for them. Data Security and Communication: - HTTPS Protocol: Ensure secure communication between the user and server. - Encryption Protocols: Implement protocols to encrypt sensitive data.
4- Communication Interface requirements	- Collect user information such as email, phone number, date of birth, and gender during the login process of the application. - Enable app permissions, such as access to the camera, microphone, sending notifications, and accessing the photo gallery.

2.5.3 Non-Functional Requirements

1- Security	Use Transport Layer Security (TLS) that encrypts data sent between the client and the server, uses digital certificates to verify the identity of the server and ensures that the data has not been tampered during transmission.
2- Capacity	Use caching techniques to improve performance and reduce the load on servers.
3- Reliability	Instagram relies on a distributed cloud infrastructure, which ensures service continuity even in the event of a server outage. Data is hosted on multiple servers so that if one of them fails, operations can be transferred to another server without affecting the user experience.
4- Maintainability	Splunk is a comprehensive application and infrastructure monitoring system that provides real-time performance monitoring of applications and servers, tracking user engagement, analyzing performance, and identifying potential issues before they impact users or the business.
5- Usability	Provide step-by-step instructions on how to use certain features in the app such as how to upload photos or how to use filters with quick and easy answers.

2.5.4 Functional Requirements

Design Requirements:

1. User Interface: A simple and intuitive design that is compatible with Instagram, featuring easy navigation menus, clear buttons, and contextual help.
2. User Experience: A smooth experience with short loading times, responsive design for different screen sizes, and customization options.

Graphics Requirements:

1. Visual Consistency: Maintain consistency with Instagram's branding in terms of colours and fonts.
2. High-Resolution Support: Ensure clarity of graphics and images across all devices.
3. Optimized Display: Smooth graphics performance without lag.

Operating System Requirements:

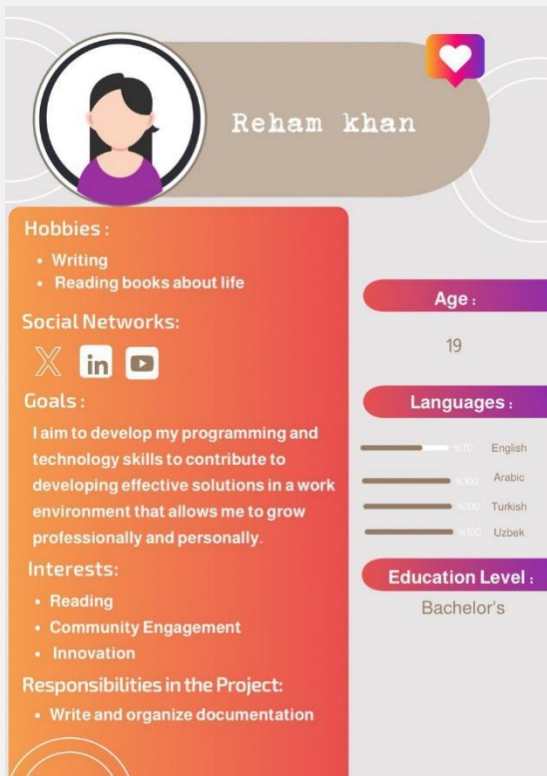
1. Android: Compatibility with Android 9.0 and above, supporting various devices.
2. iOS: Compatibility with iOS 12.0 and above, supporting multiple iPhone models.
3. Cross-System Consistency: Provide a consistent user experience across different operating systems.

Constraints:

1. Data Privacy and Security: Compliance with data protection regulations and ensuring data security.
2. Performance: Efficient use of device resources while optimizing performance.
3. Integration with Instagram: Adherence to Instagram's API and platform guidelines.
4. Scalability: Design for scalability with the growth of the user base and data volume.

2.6 Personas, Stories, and Journey Maps

2.6.1 Personas



Reham Khan

Hobbies :

- Writing
- Reading books about life

Social Networks:

X in YouTube

Goals :

I aim to develop my programming and technology skills to contribute to developing effective solutions in a work environment that allows me to grow professionally and personally.

Interests:

- Reading
- Community Engagement
- Innovation

Responsibilities in the Project:

- Write and organize documentation

Age : 19

Languages :

English: 70%
Arabic: 100%
Turkish: 100%
Uzbek: 100%

Education Level : Bachelor's



Afrah Ayoub

Hobbies :

- Reading
- Design

Social Networks:

X in YouTube

Goals :

I aim to apply my academic knowledge to make a positive impact and continuously grow personally and professionally.

Interests:

- Capturing Beautiful Moments
- Journaling
- Cooking and Trying New Recipes
- Technology

Responsibilities in the Project:

- Write and organize documentation

Age : 20

Languages :

English: 70%
Arabic: 100%

Education Level : Bachelor's



Batoul Hallaq

Hobbies :

- Drawing
- video games

Social Networks:

X Instagram

Goals :

I aim to develop my technical skills and expand my knowledge of programming and development languages.

Interests:

- Art
- anime
- video games, and everything related to technology and artificial intelligence.

Responsibilities in the Project:

- Write and organize documentation

Age : 19

Languages :

English: 70%
Arabic: 100%

Education Level : Bachelor's



Yousra Tamer

Hobbies :

- Reading

Social Networks:

X in YouTube

Goals :

I seek to apply the knowledge and skills I have gained during my academic journey in a work environment that fosters creativity and encourages growth.

Interests:

- Graphic design
- Artificial intelligence techniques
- frontend and backend developing
- communication and storytelling

Responsibilities in the Project:

- Write and organize documentation

Age : 19

Languages :

English: 80%
Arabic: 100%

Education Level : Bachelor's

2.6.2 User Stories

USER STORIES



As a mother who cares about the safety of her children on the Internet, this service will help me filter the content in Instagram, so that my children can view the appropriate content



As a university student interested in sharing my daily moments, I want The Smart Assistant to suggest post ideas based on my interests



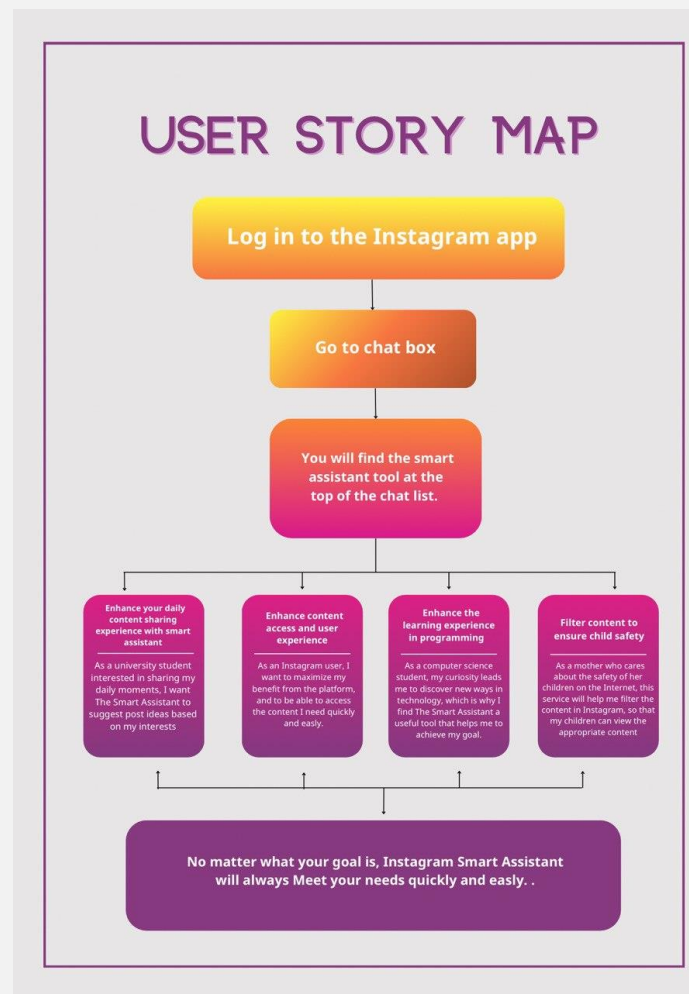
As a computer science student, my curiosity leads me to discover new ways in technology, which is why I find The Smart Assistant a useful tool that helps me to achieve my goal.



As an Instagram user, I want to maximize my benefit from the platform, and to be able to access the content I need quickly and easily.



2.6.3 User Journey Map



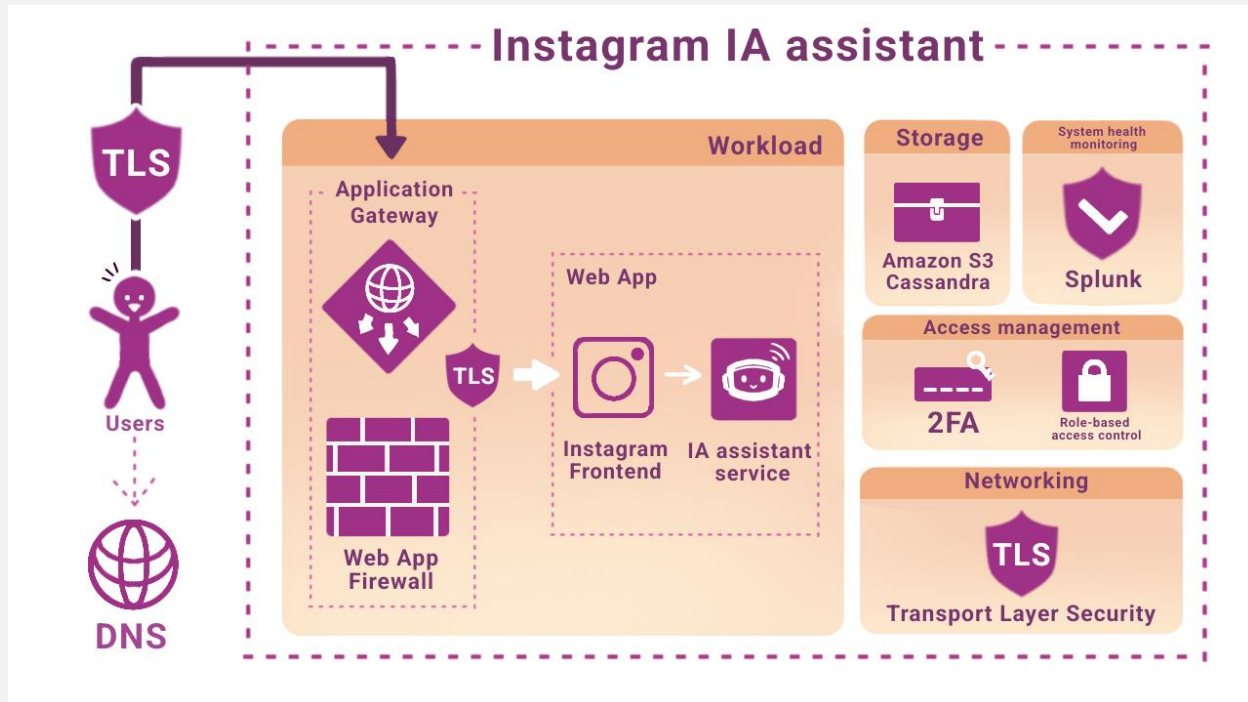
2.7 Summary

The Instagram Smart Assistant aims to enhance user engagement through features such as automatic message replies, personalized suggestions, and data analysis to provide advice on improving audience interaction. This assistant targets all Instagram users worldwide, regardless of cultural or religious backgrounds, focusing on delivering a personalized and neutral experience. The system assumes users are familiar with Instagram's functions and that smart-phones support both iOS and Android. To ensure these assumptions, the system enhances recommendation accuracy by analysing user data, ensuring a consistent experience across platforms, and developing a flexible user interface suitable for all levels, with support for multiple languages.

Core requirements include online availability 24/7 and easy access from various devices. External requirements involve a dedicated button for accessing the tool, text and voice interactions, and image recognition support. The system relies on a framework to develop an interactive interface with text analysis and machine learning services, prioritizing security through HTTPS protocols and encrypting sensitive data to protect users. Non-functional requirements emphasize security through secure connections, performance improvements using caching, reliability with a distributed infrastructure, maintenance with performance monitoring, and ease of use with simple guidance. Constraints include compliance with privacy and security regulations, efficient device resource usage, adherence to Instagram's API, and a flexible architecture that supports user expansion and data growth.

Chapter 3-Design Considerations

3.1 Software Architecture Design



- DNS: It is a system responsible for translating domain names into IP addresses that the computer can understand and communicate through.
- Web App Firewall: The firewall protects the system from external attacks.
- Amazon S3: It is a distributed cloud storage service that allows large media to be stored -securely and scalably.
- Cassandra: It is a database that manages huge amounts of data efficiently and provides the service permanently and scalably.
- Splunk: It is used to monitor the system and infrastructure and monitor the real-time performance of applications and servers.
- 2FA: It requires users to confirm their identity using an additional code sent via text messages.
- Role-based access control: This system is used to control who can access certain data or make changes to the system.
- TLS: It is a security protocol used to secure communications over the Internet and protects data exchanged between users and servers through encryption, identity verification, and data integrity.

3.2 Documentation Plan.

3.2.1 Readme and Troubleshooting

Readme:

Introduction:

The Smart Assistant in Instagram is a multifunctional tool designed to help users with various tasks such as generating ideas, answering questions, and filtering content.

Features:

1. Filtering Content to Suit Children:

Description: This feature ensures that the content shown to children is safe and appropriate.

Flow:

- The user logs in and starts a request via text or voice, specifying details such as inappropriate content or content they want to block from children.
- The system processes the request and filters the content accordingly.

2. Generating Post Ideas

Description: Helps users generate creative post ideas based on their interests and preferences.

Flow:

- The user sends a message, voice request, or attaches an image file.
- The system analyses the user's interests and generates post ideas.
- The user selects an idea and implements it.
- Post performance is tracked, and feedback is provided to the user.

3. Helping Users Find the Content They Need

Description: Assists users in quickly finding content or features within Instagram.

Flow:

- The user can either click on quick suggestions or send a request.
- The assistant processes the request and provides content suggestions based on user input.

4. Answering Technology Questions and Suggesting Technology Content

Description: Provides answers to technology-related questions and suggests relevant tech content.

Flow:

- Users can send inquiries or upload attachments for technical questions.
- The assistant processes the question and provides answers or content suggestions accordingly.

User Inputs:

Text Input: Typing messages or questions to be processed by the assistant.

Voice Input: Recording voice messages to be transcribed and processed.

Attachments: Uploading images or files to assist in generating post ideas or asking complex questions.

Quick Suggestions: Clicking predefined suggestions for faster access to certain features.

Outputs:

The assistant provides customized responses, content suggestions, and post ideas based on user input.

Content is filtered based on the target audience to ensure sensitive content is managed appropriately.

Error Detection and Troubleshooting

To ensure the best user experience, an efficient error detection mechanism is embedded in the Smart Assistant. This mechanism identifies and addresses potential issues that may arise during user interaction:

Input Parsing Failure:

The system analyses all inputs (text, voice, and attachments) to check for potential errors, such as incorrect or unintelligible entries.

If the input is not recognized, the assistant displays an interactive message to guide the user in improving the input or retrying. The user may also be prompted to review the app's permissions.

Attachments Not Uploaded Correctly:

Errors might occur due to incorrect file size or type.

The user is prompted to ensure the file size and type comply with Instagram's requirements.

Error in Request Processing:

In the case of request processing errors, such as lost server connection or system errors, the assistant alerts the user and automatically retries the request. If the issue persists, the user is directed to technical support.

Automatic Correction:

The assistant includes functions for automatically resolving common errors, such as correcting spelling mistakes or re-requesting missing information.

If automatic correction fails, the user is notified and guided on how to resolve the issue manually.

Continuous Updates:

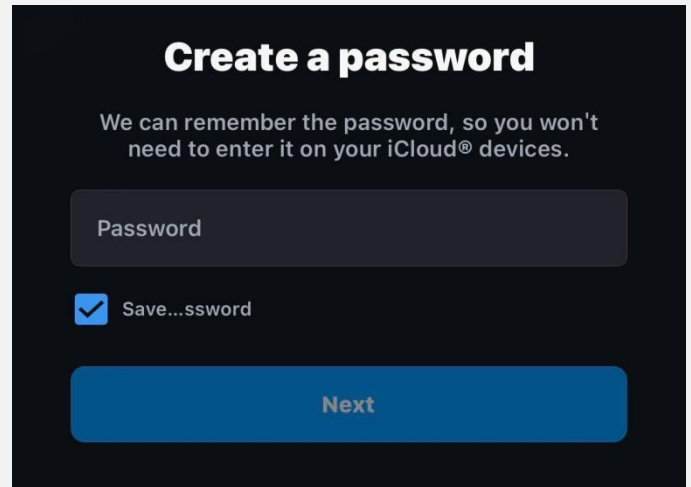
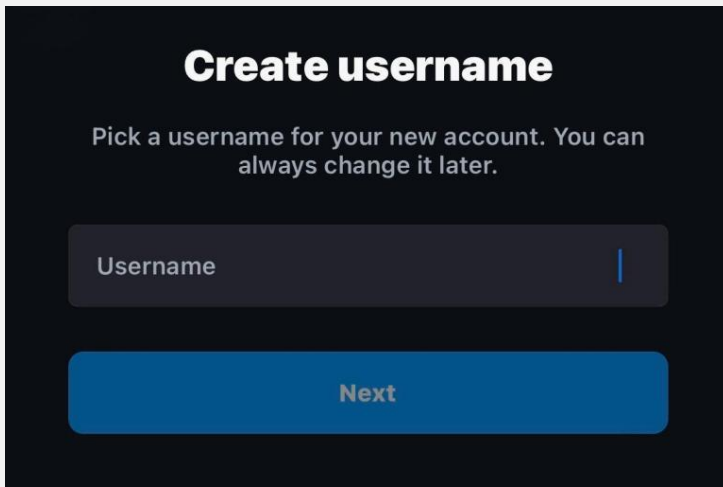
The system relies on error reports collected and analysed to enhance performance and prevent future issues. Regular updates are released to address detected problems and improve the assistant's interaction with users.

3.2.2 User Guide.

1. Sign Up and Log In

To start using the smart assistant, you must have an Instagram account. Log in to your account using your username and password.

Instagram Login Screen

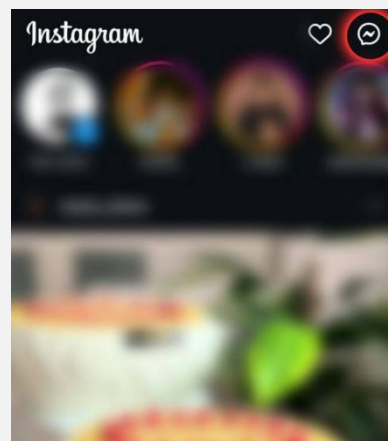
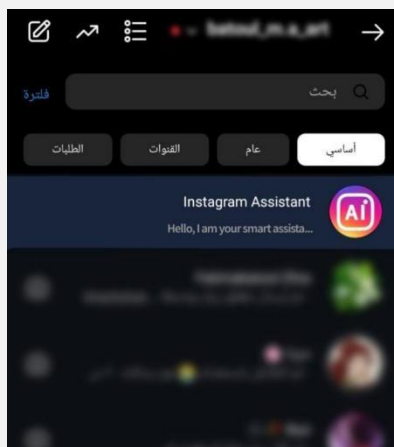


Instagram login screen prompting the user to enter their username and password.

2. Accessing the Smart Assistant

After logging in, you can access the smart assistant by tapping the Direct Messages icon in the top corner of the screen.

Instagram Direct Messages Icon

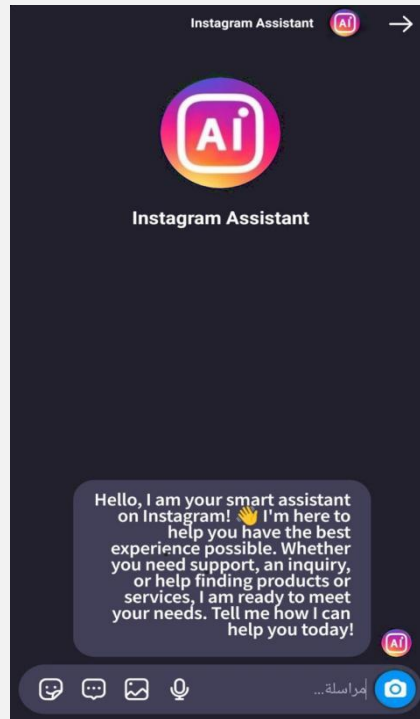


The Instagram Direct Messages icon leading to a chat with the smart assistant.

3. Using the Assistant

Start a conversation with the smart assistant by typing a message. You can request suggestions, ask questions, or ask to view specific content.

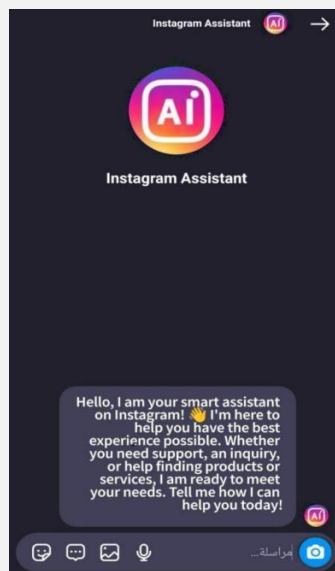
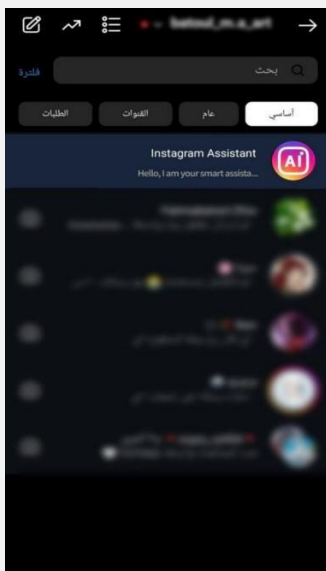
Smart Assistant Chat Interface



The chat interface with the smart assistant, where the user can send messages.

3.3 Visual content

3.3.1 Overview of User Interface



Screen Objects and Actions

The first image shows where The Smart Assistant service is inserted into the Instagram chat box, and it is in the form of a different coloured conversation at the top of the list.

The second image shows the shape of the conversation from the inside, which is a simulation of the Instagram chat design. At the bottom there is a field for writing texts and other options for sending pictures and recording voices.

3.4 Summary

This chapter outlines the software architecture design and documentation plan for the smart assistant in Instagram. The system includes tools like DNS, firewall, cloud data storage, and advanced databases to ensure data security and high performance, with security features such as two-factor authentication and access control.

The documentation includes a README file and a troubleshooting guide, showcasing features of the assistant like content filtering for children, post suggestion ideas, and support for text and voice input. It also contains a system for error monitoring and debugging.

Additionally, the user guide provides steps for registration and access to the assistant, displaying an interactive chat interface with multiple response options.

Conclusion and Future work

In conclusion, the Smart Assistant project represents a significant step forward in enhancing user experience on the Instagram platform. The Smart Assistant aims to improve user interaction with content. Initial testing has shown positive feedback from users, reflecting a genuine need for this technology.

Future Work:

1. Support for Additional Languages:

Supporting multiple languages will be a top priority in future updates. This expansion aims to enable users from all over the world to benefit from the Smart Assistant.

2. Improving AI Algorithms:

We will work on enhancing the performance of the AI algorithms to ensure high accuracy in providing suggestions. This includes using deep learning techniques and data analysis to improve the Smart Assistant's understanding of user interests and behaviours.

3. Adding New Features:

Better analysis of personal interests and the development of analytical tools to help track user preferences and provide more personalized content based on their interests.

4. Continuous Testing and Feedback:

There will be a continuous system for collecting user feedback to improve performance. Comprehensive testing will be conducted to ensure the Smart Assistant responds effectively and appropriately to user needs.

4.1 References

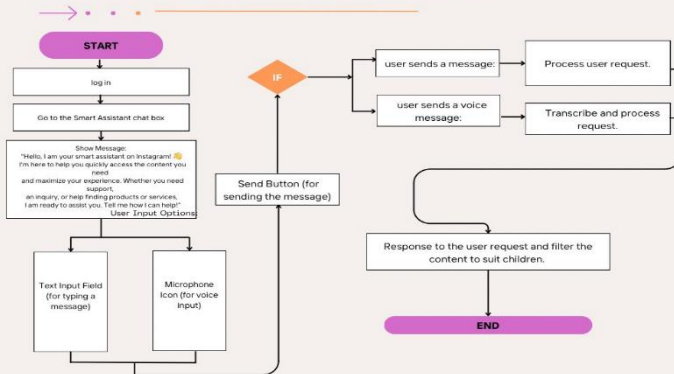
- Instagram System Design: <https://iq.opengenus.org/system-design-of-instagram/>
- Instagram Software Design: <https://techbyexample.com/instagram-system-design/>
- Instagram API: <https://developers.facebook.com/docs/instagram-platform/instagram-api-with-facebook-login>

Appendix:

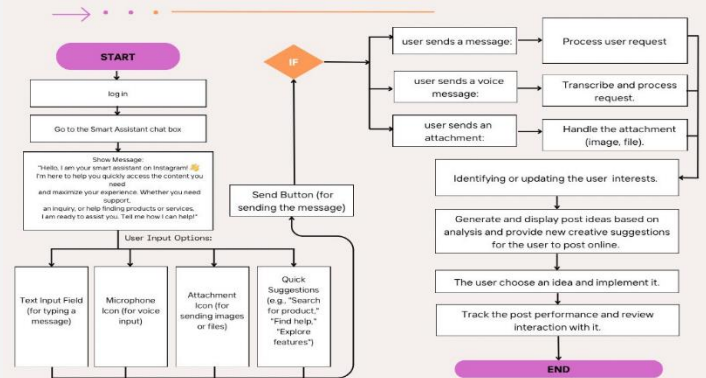
A. code sample and testing plan

a. Source Code Documentation (SCD)

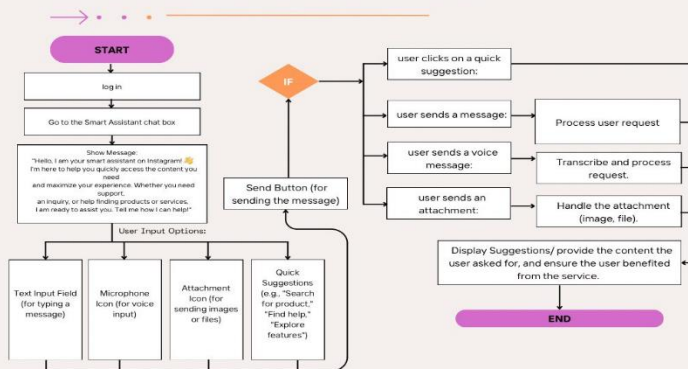
FILTERING THE CONTENT TO SUIT CHILDREN.



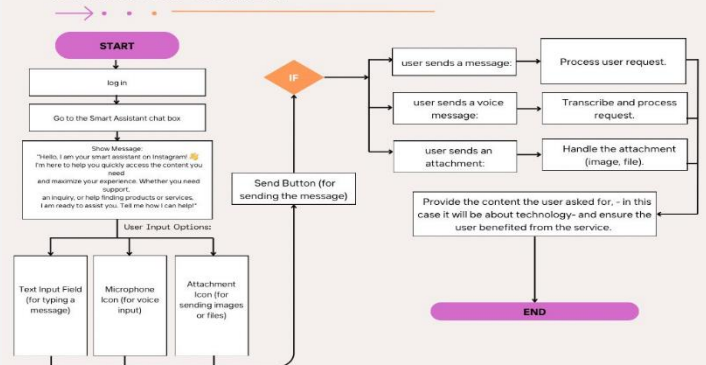
GENERATE POST IDEAS



HELP USER FINDING THE CONTENT HE NEED:



ANSWER TECHNOLOGY QUESTIONS AND SUGGEST TECHNOLOGY CONTENT



b. Quality Assurance Documentation (QAD)

• Quality Management Plan

Quality Management Plan			
Class Name:		Content Filtering for Children	
Functionality:		Filter content to ensure it is appropriate for children.	
Goal	Outcome	Target Objective	Performance Measures/Data Source(s)/ Frequency/Responsible Person
Ensure that the content complies with safety standards for children.	Content reaching children is free of inappropriate material.	Achieve accuracy in content filtering and reduce the amount of inappropriate content.	Performance Criteria/Data: Number of inappropriate content detected, number of complaints from users or parents about the content. Sources/Frequency/Person Responsible: • Sources: Content filtering tools. • Frequency: Daily testing with each content update.

Quality Management Plan			
Class Name:		Quick and Effortless Content Access	
Function:		Function:	
Functionality:		Improve the speed of accessing desired content.	
Goal	Outcome	Target Objective	Performance Measures/Data Source(s)/ Frequency/Responsible Person
Ensure that content can be accessed quickly and efficiently, reducing the effort required by users.	Deliver a smooth and fast user experience for accessing content.	Improve user engagement with the content.	Performance Criteria/Data: • User satisfaction level based on surveys and feedback. Sources/Frequency/Person Responsible: • Sources: Performance measurement tools. • Frequency: Evaluated monthly or after each major update.

Quality Management Plan			
Class Name:		Exploring New Technologies	
		Function:	
Functionality:		Assess the feasibility of applying modern technologies in the project.	
Goal	Outcome	Target Objective	Performance Measures/Data Source(s)/ Frequency/Responsible Person
Ensure the discovery and evaluation of new technologies.	Identify a set of modern technologies.	Provide a detailed report on the benefits and challenges of each technology.	Performance Criteria/Data: • Percentage of applicable technologies. • Quality and analysis of the submitted reports. Sources/Frequency/Person Responsible: • Sources: Technical publications. • Frequency: Evaluation every three months or as needed.

Quality Management Plan			
Class Name:		Suggesting Post Ideas Based on Interests	
		Function:	
Functionality:		Analyse user interests to suggest relevant post ideas.	
Goal	Outcome	Target Objective	Performance Measures/Data Source(s)/ Frequency/Responsible Person
Improve post quality by providing ideas that align with user interests to increase engagement.	Increase engagement rates with posts and achieve better interaction.	Achieve interaction with posts that are based on user interests	Performance Criteria/Data: • Analyze user satisfaction through surveys and feedback. Sources/Frequency/Person Responsible: • Sources: User analysis data. • Frequency: Weekly evaluation or after each major content campaign.

- **Test Specifications**

Test Name:	Content Filtering for Children
Description:	Enable the user to filter inappropriate content for children.
Requirement(s):	The application must allow the user to filter content to make it suitable for children.
Prerequisites:	Login and access to filtering settings.
Setup:	The user is logged in and has the ability to enable or disable filtering.

Test Name:	Quick and Effortless Content Access
Description:	Enable the user to access content quickly and with minimal effort.
Requirement(s):	The application allows the user to access content quickly and with less effort, with the option to rearrange the content if desired.
Prerequisites:	Login and availability of content to access.
Setup:	The user is logged in and has accessible content available.

Test Name:	Discovering New Methods in Technology
Description:	Enable users to discover innovative and new methods in the field of technology.
Requirement(s):	The application allows the user to explore and apply new technological methods, with the option to revert if necessary.
Prerequisites:	The user is logged in, and there are new technological methods available for discovery.
Setup:	The user is logged in and has the ability to explore the new technological methods.

Test Name:	Suggesting Post Ideas Based on Interests
Description:	Enable the user to receive post ideas based on their personal interests.
Requirement(s):	The application must allow the user to receive post suggestions that match their interests, with the option to request additional suggestions if desired.
Prerequisites:	Login and interests specified in the application.
Setup:	The user is logged in and has interests recorded in the application.

	Step	Operator Action	Expected Results	Observed Results	Pass/ Fail
	1.	Log in to the app using your email	The user should be able to log in to the app	An email input box appears, and the user can choose an existing email or create a new one	Pass
	2.	Request content filtering from the intelligent assistant	The user should be able to select the appropriate filter	Appropriate content is displayed	Pass

	Step	Operator Action	Expected Results	Observed Results	Pass/ Fail
	1.	Log in to the application using your email address.	The user should be able to log in to the application.	An email input box appears, and the user can select an existing email or create a new one.	Pass
	2.	Request new suggestions.	The user should be able to request new ideas if the initial suggestions are not suitable.	New suggestions are successfully displayed.	Pass

	Step	Operator Action	Expected Results	Observed Results	Pass/ Fail
	1.	Log in to the app using your email .	The user should be able to log in to the app	An email input box appears, and the user can choose an existing email or create a new one .	Pass
	2.	Click on the menu where the technological methods are listed.	The menu should appear, and the user should be able to select the technological method they want to explore	The menu appears with all available technological methods	Pass

	Step	Operator Action	Expected Results	Observed Results	Pass/Fail
	1.	Log in to the application using your email address.	The user should be able to log in to the application.	An email input box appears, and the user can select an existing email or create a new one.	Pass
	2.	Review suggestions.	The user should be able to review post suggestions.	Suggestions are displayed appropriately and are easy to read.	Pass

Instagram Smart Assistant (ISA)

User Manual

Version 1.0
18/10/2024

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1. Introduction

Instagram Smart Assistant (ISA)

Abbreviation: ISA

Version Number: 1.0

Release Number: 2024.1

Welcome to the Smart Assistant! The smart assistant is your new companion on Instagram, it is designed to help enhance your experience on the platform. Here, you'll find personalized tools and recommendations to make managing your account easier and more efficient.

The Instagram Smart Assistant is an innovative tool that uses artificial intelligence technologies to enhance the user experience. It also helps users manage their accounts more efficiently and monitors their account activities, which assists in understanding their interests and preferences, allowing the delivery of tailored content. With these features provided by the system, users can focus on the creative and strategic aspects of managing their accounts.

This document aims to provide a comprehensive user guide for utilizing the Instagram Smart Assistant for Chat (ISA). The guide covers all aspects, from set-up steps to advanced features, enabling users to maximize the assistant's capabilities.

This guide has been prepared based on the development phases of ISA, which included prototype design, system testing, and gathering user feedback. The main objective is to thoroughly analyse requirements to ensure continuous performance improvement.

The document is intended for Instagram users who want to enhance their interactions on the platform using the smart assistant. It can also serve as a reference for developers, product managers, and support teams interested in developing or supporting this assistant.

This guide is expected to evolve alongside the development of the smart assistant. The document will be updated regularly with each new release to include improvements and additional features, including future integrations with other Instagram features and AI updates.

As ISA deals with user data and interactions, protecting privacy and security is a top priority. The assistant follows Instagram's privacy policy, with all chat data encrypted and securely stored. No personal information is shared without the user's consent.

1.1 Overview

This system is designed to help users conduct conversations, respond to messages, and provide interaction suggestions based on pattern analysis.

Key Features:

The assistant includes features such as suggestions based on user interactions, and data analysis to provide advice on better audience engagement and controlling content display algorithms.

System Architecture:

It is based on a web architecture that operates on external servers supported by artificial intelligence technologies and natural language processing (NLP).

User Access Mode:

The smart assistant is accessed through a graphical user interface integrated into the Instagram application.

System Environment:

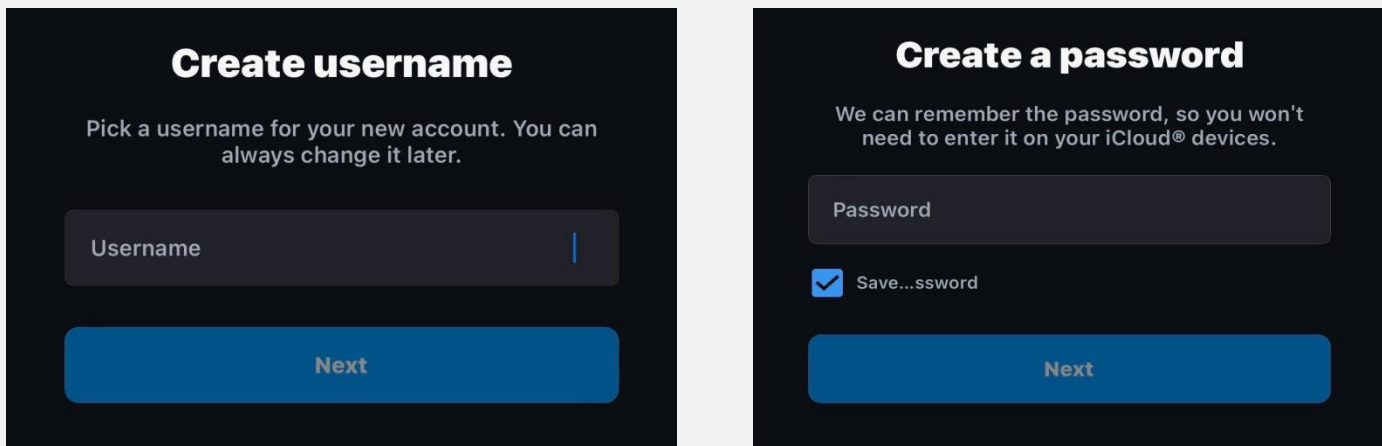
The assistant operates within the Instagram environment and can interact with all chat and communication features of the application, without requiring any special conditions or additional requirements.

2. Getting Started

2. Sign Up and Log In

To start using the smart assistant, you must have an Instagram account. Log in to your account using your user name and password.

Figure1: Instagram Login Screen



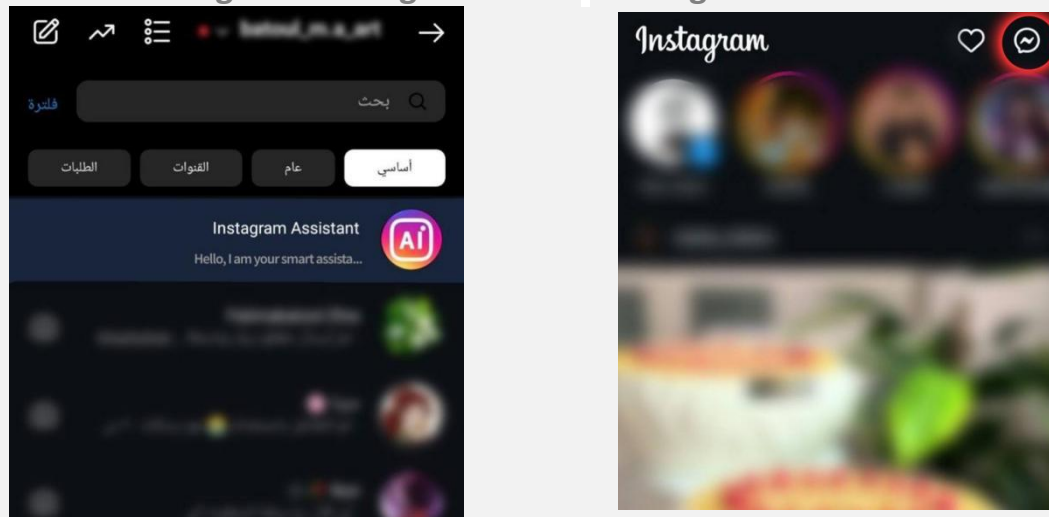
The figure displays two sequential screens from the Instagram sign-up process. The first screen, titled 'Create username', prompts the user to 'Pick a username for your new account. You can always change it later.' It features a text input field labeled 'Username' and a blue 'Next' button. The second screen, titled 'Create a password', prompts the user with 'We can remember the password, so you won't need to enter it on your iCloud® devices.' It includes a password input field, a checked checkbox labeled 'Save password', and a blue 'Next' button.

Instagram login screen prompting the user to enter their user name and password.

3. Accessing the Smart Assistant

After logging in, you can access the smart assistant by tapping the Direct Messages icon in the top corner of the screen.

Figure2: Instagram Direct Messages Icon



The Instagram Direct Messages icon leading to a chat with the smart assistant.

3. Using the Assistant

Start a conversation with the smart assistant by typing a message. You can request suggestions, ask questions, or ask to view specific content.

Figure3: Smart Assistant Chat Interface



The chat interface with the smart assistant, where the user can send messages.

2.1 Cautions & Warnings

Be cautious about the information you share with the smart assistant. It is recommended to avoid sharing sensitive information such as phone numbers, addresses, or any other personal details that may put your privacy at risk.

2.2 Set-up Considerations

Equipment and Configuration:

Supported Devices: You can use any computer, smart-phone, or tablet running a modern operating system like Windows, mac OS, Android, or iOS.

Input Devices:

Keyboard: Used to type messages and input text.

Touch-screen: If using a mobile device, you can interact with the assistant through touch.

Voice: You can use a microphone to interact with the smart assistant.

Output Devices:

Screen: Displays messages and interactions with the smart assistant.

Network Configuration:

Internet Connection: You'll need an internet connection to use the smart assistant, either via Wi-Fi or mobile data.

Servers: The smart assistant service relies on servers that host the Instagram app and provide data processing. When you use the assistant, your queries are sent to Instagram's servers, where they are analysed, and results are returned.

2.3 System Organization & Navigation

1. Home Page

When you open the Instagram app, you'll see the home page displaying posts and content from the accounts you follow. The page also has a tool-bar at the bottom that allows quick access to the home, search, notifications, and your profile.

Navigation: To access the smart assistant, tap on the Direct Messages icon in the upper right corner. From there, you can start a new conversation with the assistant.

2. Smart Assistant

After entering the messages section, the smart assistant will appear as a new chat, allowing for direct and easy interaction.

Navigation:

- You can start a new conversation with the assistant by clicking on its name.
- Use the chat box to type your queries or commands, and the assistant will respond based on your requests.

2.4 Exiting the System

Ending the Conversation: You can easily end the conversation by clicking the "Back" button or returning to the main messages' menu.

3. Using the System

Content Filtering Feature:

You can request content filtering based on specific criteria, such as keywords or content type (photos, videos, text), by giving a command to the smart assistant to perform the action.

Required Inputs: Simply specify the keywords or criteria you wish to use.

Expected Outputs: The smart assistant will block posts that match the criteria you've set.

Requesting Customized Content:

You can request to view personalized content according to your preferences and interests.

Required Inputs: Specify the type of content you want to see.

Expected Outputs: The smart assistant will display content that aligns with the preferences you've provided.

Requesting Information About Your Target Audience:

You can request information about your target audience, such as age, groups, geographical locations, or interests.

Required Inputs: Specify the type of information you'd like to know, such as "most engaged follower interests" or "demographic analysis of followers."

Expected Output: The smart assistant will provide you with a detailed report containing an analysis of your target audience based on available data.

Appendix A: Glossary

Table 1 - Glossary

Term	Acronym	Definition
<i>Instagram Smart Assistant</i>	<i>ISA</i>	<i>An AI-powered tool integrated within Instagram that helps users manage their accounts by providing personalized recommendations and automating certain interactions.</i>
<i>Artificial Intelligence</i>	<i>AI</i>	<i>A branch of computer science that simulates human intelligence in machines, allowing ISA to analyse data and offer tailored suggestions.</i>
<i>Natural Language Processing</i>	<i>NLP</i>	<i>A technology that enables computers to understand, interpret, and respond to human language, allowing ISA to interact with users via text and voice inputs.</i>
<i>Graphical User Interface</i>	<i>GUI</i>	<i>A visual way for users to interact with ISA through buttons, icons, and other graphical elements within the Instagram app.</i>
<i>Pattern Analysis:</i>	-	<i>A method used by ISA to analyse user behaviour and interactions over time to provide relevant content suggestions and automate responses.</i>
<i>Content Display Algorithms:</i>	-	<i>Systems that determine the order and priority of content shown to Instagram users, which ISA can influence based on user preferences and interactions.</i>
<i>Web Architecture:</i>	-	<i>The infrastructure that enables ISA to operate using external servers and handle large amounts of data.</i>
<i>Servers:</i>	-	<i>Powerful computers that store data and run services like ISA. When a user interacts with the assistant, their data is processed on Instagram's servers.</i>
<i>Encrypted:</i>	-	<i>A method of securing data by converting it into a code to prevent unauthorized access, ensuring user privacy in ISA's communications.</i>

Term	Acronym	Definition
<i>Input Devices:</i>	-	<i>Devices used to interact with ISA, such as a keyboard, touch-screen, or microphone.</i>
<i>Output Devices:</i>	-	<i>The screen where ISA displays responses and user interactions.</i>
<i>Tailored Content:</i>	-	<i>Customized recommendations and interactions provided by ISA based on user preferences and past behaviors.</i>
<i>Prototype Design:</i>	-	<i>An early version of ISA that was tested during development to ensure functionality and gather user feedback for improvements.</i>
<i>User Access Mode:</i>	-	<i>The method by which users interact with ISA, either through text, voice commands, or other input devices integrated within Instagram's interface.</i>

Appendix B: Referenced Documents

This section summarizes the relationship between this document and other relevant documents. The following table provides identifying information for all documents used to develop this document, including relevant technical documentation, as well as other documents that were essential in the development of the Smart Assistant feature for Instagram.

Table 2 - Referenced Documents

Document Name	Document Location and/or URL	Issuance Date
Project Report	<i>Go To Document</i>	1810/2024